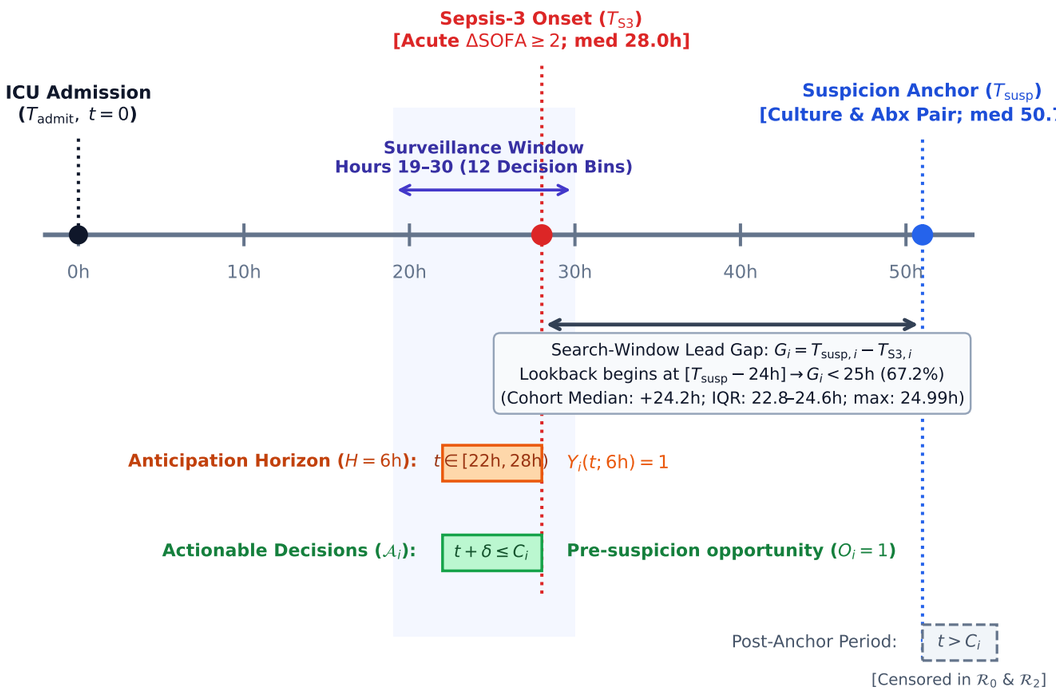
A. Operational Clocks & Actionable Decision Boundary



Mathematical Risk Set Formulation (§2.3):

- Evaluated Base Set: $\mathcal{R} = \{(i,t) : i \in \mathcal{C},\ t \in \{19 \dots 30\}\text{h},\ t < T_{S3,i}\}$ (97,700 windows across 8,254 stays)
- Anticipation Label: $Y_i(t;H) = \mathbf{1}\{t < T_{S3,i} \leq t + H\}$ ($H = 6$h; 678 positive / 97,022 negative, $\pi = 0.694\%$)
- Exploratory Hybrid: $\mathcal{R}_\delta^{\text{hybrid}} = \{(i,t) \in \mathcal{R} : t + \delta \leq C_i\}$ ($\mathcal{R}_0$: 70,192; $\mathcal{R}_2$: 70,129; T_{susp} cases, T_{Rx} controls)
- Uniform Earliest-Rx: $\mathcal{R}_\delta^{\text{Rx}} = \{(i,t) \in \mathcal{R} : t + \delta \leq T_{Rx,start,i}\}$ ($N_+ = 0$; prevalence 0.000%; discrimination not estimable)

B. Population Opportunity Attrition

